MID-SEMESTER RESEARCH PROJECT MILESTONE: ONE-PAGE SUMMARY

Student: Opol Charis | **Registration No.:** S24B23/003 | **Access No.:** B29654 | **Programme:** BSc Computer Science

Title: Rule-Constrained Reinforcement Learning for Ambiguity-Aware IT Incident Triage

Problem and research objective

Problem. When an IT incident is reported, staff must act before they know the cause, and the first evidence is often incomplete or contradictory. They can investigate further, try a likely fix, or hand the incident to a more senior person. AI tools for IT operations (AIOps) give one answer per ticket: they cannot ask for more evidence, signal that they are unsure, or check their suggestion against the organisation's rules [1].

Objective. To build and evaluate an AI agent that learns when to investigate further, when to recommend a fix the rules allow, and when to escalate the incident to a human expert.

What the research shows, the gap and my contribution

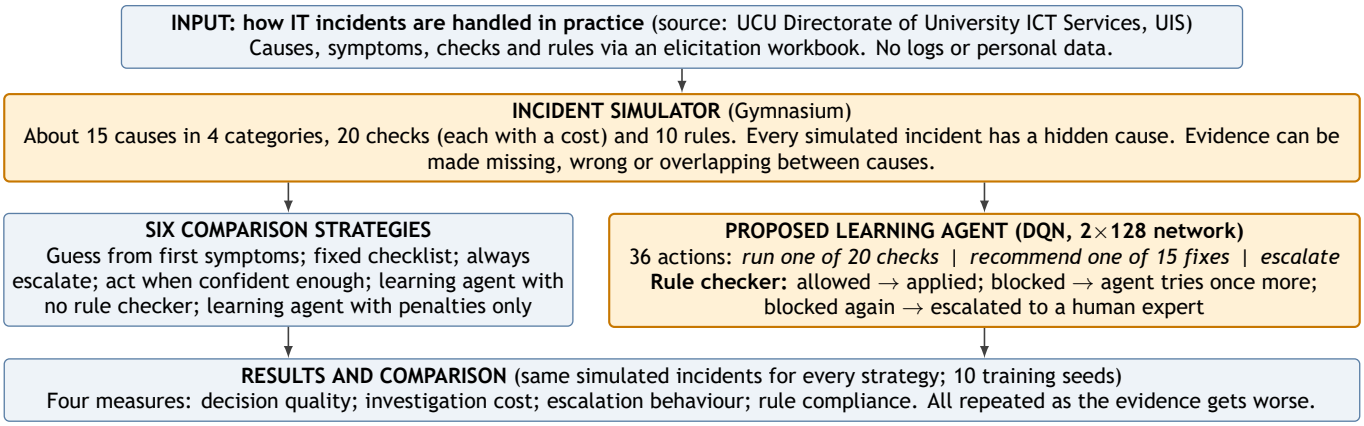
Systematic review (PRISMA 2020; Scopus, IEEE Xplore, ACM Digital Library; 4,399 records screened down to **34 included studies**) [4]:

- 17 of the 34 studies let the agent gather information, but only 7 include escalation to a person, as in learning to defer [3]. **Only one combines all three choices**, and there escalation is fixed by design, not learned.
- Rule enforcement is mostly a reward penalty, which does not stop rule breaks [2]. Only 5 studies use a hard rule filter ("shield"), and only 3 say what happens to a blocked decision.
- Decision quality is measured in 31 studies; escalation in only 5 and rule compliance in 4.

Gap. No reported agent *learns* to investigate, act or escalate for IT service incidents under explicit rules, or is evaluated on all four measures as the evidence gets worse.

Contribution. (1) An IT incident simulator built from UCU practice, with adjustable evidence quality. (2) An agent with all three choices and a separate rule checker. (3) A comparison with six strategies on four measures.

Full research pipeline



Progress to date

Done (evidence in brackets)

- **Approval:** UIS has approved the study and will supply the incident scenarios.
- **Stakeholder interview** (19 Oct) with a UIS senior developer informed the design: application-led categories, API logs as evidence, escalation to a human expert [findings note].
- **Systematic review:** 34 studies, gap confirmed [23-page draft]; **Chapters 1-3** drafted [42 pages].
- **Elicitation workbook** ready: 15 causes, 20 checks, 10 rules [workbook].
- **Environment:** construction started.
- **Implementation plan** written [2-page concept note].

Not yet done. Elicitation session, training and experiments, so no results are claimed.

Next five weeks

W2 (26 Oct): elicitation with UIS; finish the simulator; rule checker; first baselines.
W3 (2 Nov): UIS staff review sample incidents; model frozen; train the agent (10 seeds).
W4 (9 Nov): compare with the six strategies; ablations.
W5 (16 Nov): test with worse evidence; analyse.
W6 (23 Nov): prototype, poster and Chapter 4.

Prototype. The simulator and trained agent, with a screen that replays an incident step by step beside the comparison results.
Evaluation. The four measures above, mean of 10 seeds with 95% confidence intervals.

Key references

[1] Chen, Z., et al. (2020). Towards intelligent incident management: Why we need it and how we make it. *Proc. ACM ESEC/FSE*, 1487–1497. [2] Alshiekh, M., Bloem, R., Ehlers, R., Könighofer, B., Niekum, S., & Topcu, U. (2018). Safe reinforcement learning via shielding. *Proc. AAAI*, 32(1). [3] Madras, D., Pitassi, T., & Zemel, R. (2018). Predict responsibly: Improving fairness and accuracy by learning to defer. *NeurIPS*, 31. [4] Opol, C. (2026). Reinforcement learning for investigating, acting and escalating in IT incident triage and response: A systematic literature review. Draft, Uganda Christian University.